Unlike the case of Riemannian geometry, there is no ready-made geometrical opus sitting on some mathematician's shelf that string theorists can adopt and put in the service of quantum geometry. Instead, physicists and mathematicians are now vigorously studying string theory and, little by little, piecing together a new branch of physics and mathematics. Although the full story has yet to be written, these investigations have already uncovered many new geometrical properties of spacetime entailed by string theory—properties that would almost certainly have thrilled even Einstein.

The Heart of Riemannian Geometry

If you jump on a trampoline, the weight of your body causes it to warp by stretching its elastic fibers. This stretching is most severe right under your body and becomes less noticeable toward the trampoline's edge. You can see this clearly if a familiar image such as the Mona Lisa is painted on the trampoline. When the trampoline is not supporting any weight, the Mona Lisa looks normal. But when you stand on the trampoline, the image of the Mona Lisa becomes distorted, especially the part directly under your body, as illustrated in Figure 10.1.

This example cuts to the heart of Riemann's mathematical framework for describing warped shapes. Riemann, building on earlier insights of the mathematicians Carl Friedrich Gauss, Nikolai Lobachevsky, Janos Bolyai, and others, showed that a careful analysis of the distances between all locations on or in an object provides a means of quantifying the extent of its curvature. Roughly speaking, the greater the (nonuniform) stretching—the greater the deviation from the distance relations on a flat shape—the greater the curvature of the object. For example, the trampoline is most significantly stretched right under your body and therefore the distance relations between points in this area are most severely distorted. This region of the trampoline, therefore, has the largest amount of curvature, in line with what you expect, since this is where the Mona Lisa suffers the greatest distortion, yielding the hint of a grimace at the corner of her customary enigmatic smile.

Einstein adopted Riemann's mathematical discoveries by giving them a precise physical interpretation. He showed, as we discussed in Chapter 3, that the curvature of spacetime embodies the gravitational